LIÈGE université
Sciences Appliquées

Synchronous generators

ELEC0447 - Analysis of electric power and energy systems

Bertrand Cornélusse

September 7, 2026

Overview

1. Operating principle
2. Types of synchronous machines
3. Equivalent circuit and modeling
4. Properties and reactive power control
5. Generators in power flow analysis

What will we learn today?

- What synchronous generators are and how they work.
- How we can model them for power system analyses.
- How to include generator capability limits in power flow computations.

You will be able to solve exercises 9.5 to 9.9 of Ned Mohan's book:

Mohan, Ned. *Electric power systems: a first course*. John Wiley & Sons, 2012.

Operating principle

Synchronous machines

Synchronous machines produce the major part of electrical energy worldwide:

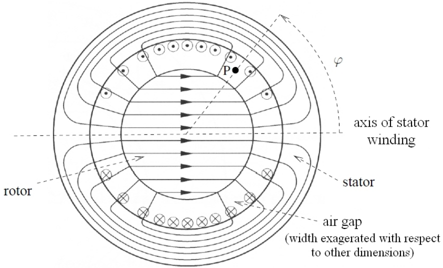
- Unit ratings range from a few kVA (emergency generators) to a few hundred MVA, with the largest units exceeding 1500 MVA (nuclear power plants).

Beyond bulk active power production, they play three critical system roles:

- **Frequency setting**: They impose the synchronous frequency of grid voltages and currents.
- **Energy buffer**: They store significant kinetic energy in their rotating masses (providing instantaneous system inertia).
- **Voltage control**: They can produce or consume reactive power by adjusting their rotor field current.

Magnetic field created by the stator

- **Stator (armature)**: Stationary outer structure separated from the rotor by a small air gap.
- Subjected to time-varying magnetic flux $\implies$ built from thin steel laminations to minimize **eddy (Foucault) currents**.
- Equipped with three windings (a, b, c) distributed $120°$ apart in space.
- Magnetic field lines cross the air gap in a radial direction.

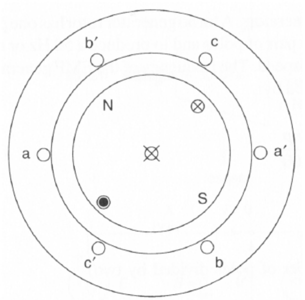


Field from direct current in one stator winding

Three-phase stator winding layout

The amplitude $B(\varphi)$ of the flux density along the air gap:

- Is a periodic function of spatial angle φ with period 2π.
- Has a discrete "staircase" shape in practice due to discrete slots.
- Is shaped as close as possible to a pure sinusoid by carefully distributing conductor coils along stator slots.



Three-phase winding layout (single turn shown)

Rotating magnetic field I

Total magnetic flux density created by the three phases at spatial angle φ:

$$B_{3\phi}(\varphi) = ki_a \cos\varphi + ki_b \cos\left(\varphi - \frac{2\pi}{3}\right) + ki_c \cos\left(\varphi - \frac{4\pi}{3}\right)$$

When balanced three-phase sinusoidal currents flow in the windings:

$$i_a(t) = \sqrt{2}I\cos(\omega t + \psi)$$
$$i_b(t) = \sqrt{2}I\cos\left(\omega t + \psi - \frac{2\pi}{3}\right), \quad i_c(t) = \sqrt{2}I\cos\left(\omega t + \psi - \frac{4\pi}{3}\right)$$

Rotating magnetic field II

Using trigonometric identities:

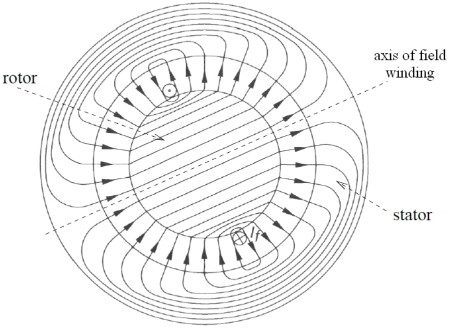
$$B_{3\phi}(\varphi, t) = \frac{3\sqrt{2}kI}{2}\cos(\omega t + \psi - \varphi)$$

Key conclusion

The three-phase AC currents together produce a magnetic field of constant amplitude rotating at electrical angular speed $\omega = 2\pi f$ (Ferraris theorem).

Magnetic field created by the rotor

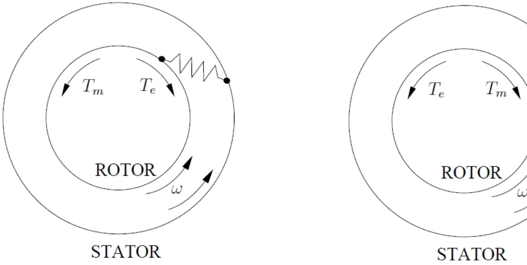
- **Rotor**: The rotating cylindrical or salient part mounted on the turbine shaft.
- In steady state, a direct current (I_f) flows through the rotor winding.
- This winding is known as the **field winding** (*enroulement d'excitation*).
- The DC current creates a magnetic field with alternating North and South poles that rotates along with the rotor.



Magnetic field produced by rotor field winding

Interaction between fields & electromechanical conversion

- In steady state, the rotor rotates at the exact same angular speed ω as the stator magnetic field: **synchronism**.
- Therefore, the stator and rotor magnetic fields are stationary with respect to each other.
- Both fields tend to align in the manner of two coupled magnets.
- Displacing them creates an electromagnetic torque T_e.
- There exists a maximum torque $T_{e,\max}$ beyond which synchronism is lost.



Mechanical analogy: generator mode $\leftrightarrow$ motor mode

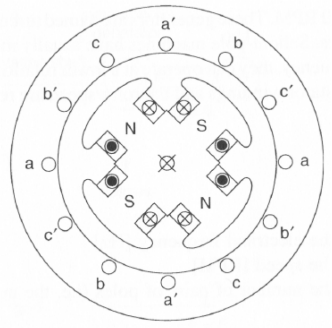
Types of synchronous machines

Machines with multiple pole pairs

Certain prime movers (e.g. hydro turbines) run at low rotational speed, yet the electrical frequency must remain $f = 50$ Hz ($T = 20$ ms):

- The rotor is equipped with p pairs of poles ($2p$ poles).
- In one mechanical revolution, each stator winding experiences p electrical cycles.
- Rotational speed n (in rpm) is linked to grid frequency f:

$$n = \frac{60f}{p} = \frac{3000}{p}\ \text{rpm} \quad \text{(at 50 Hz)}$$

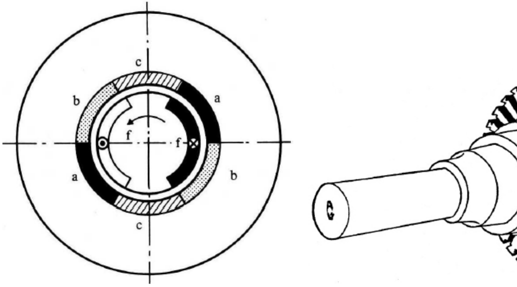


Example: $p = 2$ (4 poles, $n = 1500$ rpm)

Round-rotor generators (turbo-alternators)

Driven by high-speed steam or gas turbines:

- Typically $p = 1$ (3000 rpm) or $p = 2$ (1500 rpm in nuclear plants).
- Cylindrical rotor forged from solid steel alloy; small diameter and long length (due to severe centrifugal forces).
- Field winding embedded in milled rotor slots.
- Extreme power density: cooled by pressurized hydrogen ($7\times$ higher thermal conductivity than air) or demineralized water in hollow conductors.

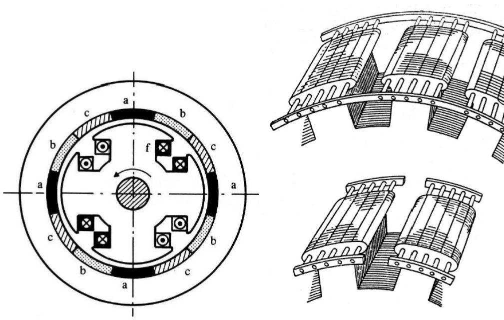


Cross-section of a round-rotor machine

Salient-pole generators

Driven by low-speed hydraulic turbines or diesel engines:

- Number of pole pairs is large ($p \geq 4$, often $p \in [10, 40]$).
- Protruding (*salient*) poles with concentrated field coils mounted on a large rotor wheel.
- Large diameter, short axial length.
- Rotor poles are made of laminated steel sheets.
- Usually cooled by natural or forced airflow around the rotor.



Cross-section of a salient-pole machine

Damper windings (*amortisseurs*)

- Conductive bars (copper/brass) placed in rotor pole shoes and short-circuited at ends with rings (similar to a squirrel cage).
- **In steady state**: The stator and rotor magnetic fields rotate synchronously $\implies$ no relative motion, zero induced current in the dampers.
- **During a disturbance**: The rotor accelerates or decelerates relative to the synchronous magnetic field $\implies$ currents are induced in the damper bars.
- According to Lenz's law, these induced currents produce a **damping torque** that counteracts rotor oscillations and helps restore synchronism.
- In solid round-rotor machines, eddy currents flowing directly in the rotor steel provide a similar intrinsic damping effect.

Equivalent circuit and modeling

Induced electromotive force (emf)

Two separate magnetic sources induce voltages in the stator windings:

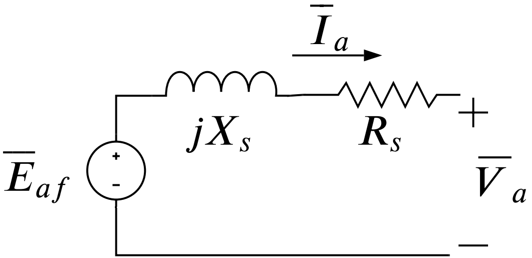
1. **Excitation field**: The rotating rotor field induces an internal electromotive force (emf) denoted $\bar{E}_{af}$ (or open-circuit voltage).
2. **Armature reaction**: The AC currents flowing in the stator windings produce their own magnetic field, inducing a voltage drop modeled as $-jX_s\bar{I}_a$.

Assuming no magnetic saturation, the resulting terminal phase voltage $\bar{V}_a$ is obtained by superposition:

$$\bar{V}_a = \bar{E}_{af} - R_s\bar{I}_a - jX_s\bar{I}_a$$

where R_s is the stator winding resistance and X_s is the **synchronous reactance**.

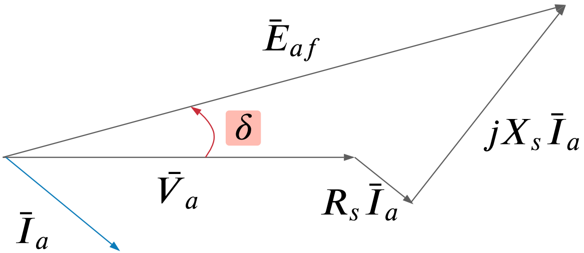
Per-phase equivalent circuit



$$\bar{V}_a = \bar{E}_{af} - (R_s + jX_s)\bar{I}_a$$

- R_s: Stator resistance.
- X_s: Synchronous reactance (leakage + magnetizing reactance).
- δ: Angle between $\bar{E}_{af}$ and $\bar{V}_a$, called the **internal angle** or **load angle**.

Phasor diagram



- $\bar{V}_a = V_a\angle 0^\circ$ is chosen as the reference.
- $\bar{I}_a$ is the stator phase current with phase lag ϕ.
- Resistive drop $R_s\bar{I}_a$ is parallel to $\bar{I}_a$.
- Reactive drop $jX_s\bar{I}_a$ leads $\bar{I}_a$ by 90°.
- Vector sum yields the internal emf:

$$\bar{E}_{af} = \bar{V}_a + R_s\bar{I}_a + jX_s\bar{I}_a$$

Parameters and transient reactances

Typical machine parameters (in per unit on machine rating base):

- Stator resistance: $R_s \approx 0.005$ pu (frequently neglected in grid studies).
- Synchronous reactance: $X_s \in [1.5, 2.5]$ pu (for round-rotor turbo-generators).

Machine models for transient conditions: Depending on the time frame of the phenomenon, stator reaction is hindered by rotor screening circuits:

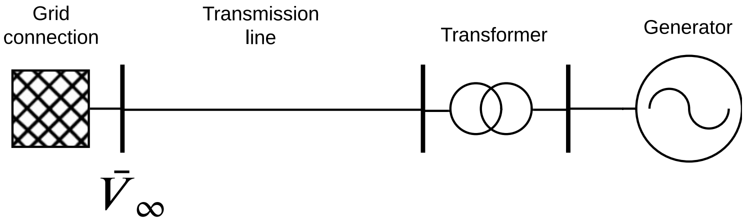
- **Subtransient state** (first few cycles after a short circuit): $X_s'' \ll X_s$ $(0.15 - 0.25$ pu).
- **Transient state** (hundreds of milliseconds): $X_s' < X_s$ $(0.25 - 0.40$ pu).
- **Steady state**: X_s $(1.5 - 2.5$ pu).

$$\implies X_s'' < X_s' < X_s$$

Properties and reactive power control

Power transfer to an external grid I

Consider a generator connected to an infinite bus ($\bar{V}_\infty = V_\infty \angle 0^\circ$) through a step-up transformer and transmission line:



Power transfer to an external grid II

Neglecting resistances, let $X_T = X_s + X_{tr} + X_{line}$ be the total reactance between $\bar{E}_{af}$ and $\bar{V}_{\infty}$. The three-phase active power delivered to the grid is:

$$P = \frac{E_{af} V_{\infty}}{X_T} \sin\delta$$

where δ is the angle between internal emf $\bar{E}_{af}$ and infinite bus voltage $\bar{V}_{\infty}$.

Steady-state stability limit

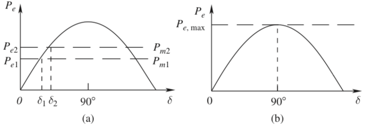


FIGURE 9.12 Steady state stability limit.

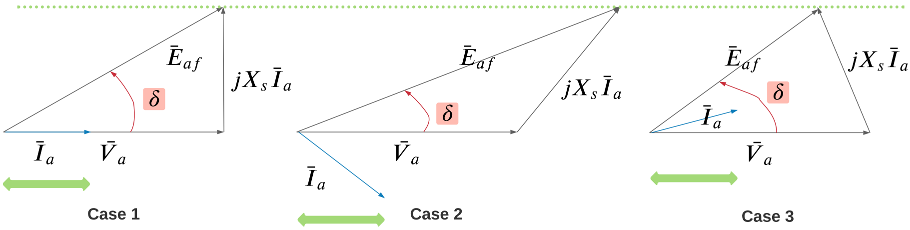
- When excitation E_{af} is constant, active power P is strictly proportional to $\sin\delta$.
- Maximum power is reached at $\delta = 90°$:

$$P_{\max} = \frac{E_{af}V_\infty}{X_T}$$

- For $\delta > 90°$, $\frac{dP}{d\delta} < 0$: additional mechanical power causes rotor acceleration, increasing δ and reducing electrical power $\implies$ loss of synchronism.
- In operation, δ is kept well below $90°$, typically $\delta \leq 40°$.

Adjusting reactive power via field current

Suppose the generator delivers constant active power P, requiring $E_{af} \sin \delta = \text{cst}$ and $I_a \cos \phi = \text{cst}$:



By varying the DC field excitation current I_f, we directly control the amplitude E_{af} and the reactive power Q exchanged with the grid.

Overexcitation vs. Underexcitation

Overexcitation ($I_f \nearrow$, $E_{af} \nearrow$):

- Current $\bar{I}_a$ lags terminal voltage $\bar{V}_a$ ($\phi > 0$).
- Generator **supplies** reactive power to the grid:

$$Q = 3V_aI_a \sin\phi > 0$$

- The generator behaves as a **capacitor** connected to the power system.

Underexcitation ($I_f \searrow$, $E_{af} \searrow$):

- Current $\bar{I}_a$ leads terminal voltage $\bar{V}_a$ ($\phi < 0$).
- Generator **absorbs** reactive power from the grid:

$$Q = 3V_aI_a \sin\phi < 0$$

- The generator behaves as an **inductor** connected to the power system.

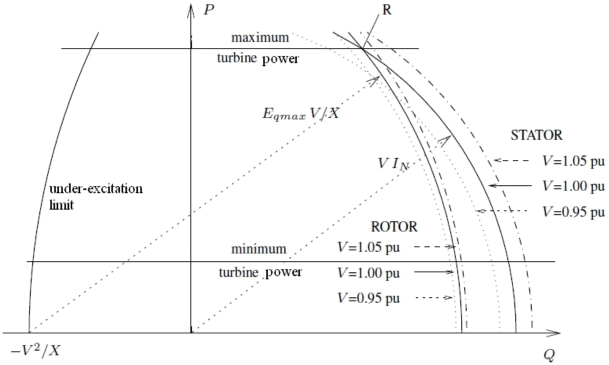
Voltage regulation and synchronous condensers

- **Automatic Voltage Regulation (AVR):**
  - Controls the DC rotor excitation current I_f via feedback control.
  - Dynamically adjusts internal emf E_{af} and reactive power Q to regulate the terminal bus voltage to a target setpoint V_0.

- **Synchronous Condensers:**
  - Synchronous machines running without a mechanical load (drawing only enough active power to cover internal losses).
  - Used exclusively to supply or absorb continuously variable reactive power and provide short-circuit power and inertia to the grid.

Generator capability curve

The capability curve defines the permissible operating region in the (P, Q) plane under constant terminal voltage V:



Operating limits breakdown

1. **Prime mover (turbine) limit**: Maximum mechanical power input ($P \leq P_{\max}$). A minimum power $P_{\min}$ also exists for flame stability in thermal boilers.
2. **Stator thermal limit**: Armature Joule losses ($R_s I_a^2$). Current magnitude is capped ($I_a \leq I_{a,\max}$):

$$P^2 + Q^2 \leq (V I_{a,\max})^2 \quad \text{(circle centered at origin)}$$

3. **Rotor thermal limit**: Joule losses in field winding ($R_f I_f^2$). Maximum excitation voltage $E_{af} \leq E_{af,\max}$:

$$P^2 + \left(Q + \frac{V^2}{X_s}\right)^2 \leq \left(\frac{V E_{af,\max}}{X_s}\right)^2 \quad \text{(circle centered at } (0, -V^2/X_s)\text{)}$$

4. **Underexcitation / Stability limit**: Minimum field excitation required to avoid excessive internal angle δ and loss of synchronism.

Generators in power flow analysis

Inclusion in power flow: PV to PQ bus switching

In power flow formulations:

- A synchronous generator is initially modeled as a **PV bus**:
  - Active power P is fixed by turbine dispatch.
  - Voltage magnitude $|V|$ is fixed by the AVR setpoint.
  - Reactive power Q and phase angle θ are unknown.
- After computing power flow equations, the required reactive power Q is checked against capability limits:

$$Q_{\min} \leq Q \leq Q_{\max}$$

- If a limit is violated (e.g. $Q > Q_{\max}$):
  - The generator cannot provide enough field current to sustain the voltage setpoint.
  - The bus is switched from a **PV bus** to a **PQ bus** with $Q = Q_{\max}$ (or $Q_{\min}$).
  - Voltage $|V|$ is freed and becomes a state variable.
  - Power flow is re-solved iteratively until no limits are violated.

Simulation with PandaPower

In PandaPower, generator limits can be directly enforced during AC power flow calculations:

```
pp.runpp(net, enforce_q_limits=True)
```

Google Colab Notebook

Explore synchronous machine modeling and PV/PQ switching in PandaPower:
https://colab.research.google.com/drive/1Q9E62wzrKiCsBIyRH9JnCTMSibSlY9OG?usp=sharing

References

- Course notes of ELEC0014 (*Electric Energy Systems*) by Pr. Thierry Van Cutsem, University of Liège.
- Mohan, Ned. *Electric power systems: a first course*. John Wiley & Sons, 2012.
- L. Thurner, A. Scheidler, F. Schäfer et al., "pandapower - an Open Source Python Tool for Convenient Modeling, Analysis and Optimization of Electric Power Systems", *IEEE Transactions on Power Systems*, vol. 33, no. 6, pp. 6510-6521, Nov. 2018.